

Nursing Implications in the Administration of Paclitaxel (TAXOL®)

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Paclitaxel is an antimicrotubule agent that was recently approved by the Food and Drug Administration (FDA). Because it is insoluble in water, it is mixed with Cremophor® EL (polyoxyethylated castor oil). The addition of Cremophor requires the use of glass bottles or polyolefin or polypropylene bags and polypropylene-lined tubings for the administration of paclitaxel. In addition, 0.22 µm in-line filters are necessary. The potential side effects of paclitaxel during its infusion include hypersensitivity reaction (HSR), cardiotoxicity, infiltration, diarrhea, and nausea, whereas myelosuppression, neurotoxicity, myalgias, arthralgias, alopecia, fatigue, headache, taste changes, and minor alterations in renal and liver function tests can occur after its administration. Premedications including an H₂ blocker, a corticosteroid, and an antihistamine are necessary to minimize the occurrences of HSRs. Vital signs should be monitored throughout the infusion to assess for HSRs and cardiotoxicity. Paclitaxel has been shown to be safe in both the inpatient and outpatient settings. Nursing care includes the administration of the drug, the assessment and management of side effects, and psychosocial support of patients receiving the drug.

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PACLITAXEL (TAXOL® [Bristol-Myers Squibb Co., Princeton, NJ]) is an antimicrotubule agent that has become the latest anticancer drug to be approved by the US Food and Drug Administration (FDA). Nurses have an important role in the administration of this new drug, as well as in the prevention and management of side effects. In addition, nurses assist patients in coping with paclitaxel treatment as well as its discontinuation. This article will describe the qualities of paclitaxel that influence nursing care of patients receiving this drug and will outline the specific nursing actions necessary to prevent and manage potential side effects during the infusion.

Like other antimicrotubule agents, the structure of paclitaxel is complex. However, paclitaxel differs from other antimicrotubule agents such as colchicine and the vinca alkaloids in that, rather than inducing microtubule disassembly, paclitaxel promotes the assembly of microtubules and stabilizes their formation.¹

Paclitaxel is insoluble in water; therefore, the addition of 50% Cremophor® EL (polyoxyethylated castor oil) and 50% dehydrated alcohol is

required in its formulation. Cremophor used singularly has been associated with toxicity, and at high doses it has caused vasodilation, labored breathing, lethargy, hypotension, and death in dogs; however, no cumulative effects have been noted.² In addition, leaching of the plasticizer di-(2-ethylhexyl) phthalate (DEHP) occurs when Cremophor comes into contact with polyvinyl chloride bags. Because DEHP is hepatotoxic, exposure should be minimized. Dilution of paclitaxel in intravenous (IV) fluids produces slight haze and a few minute particles. Specific equipment is necessary for the administration of paclitaxel to minimize patient exposure to both DEHP and the minute particles found in the solution.³

Paclitaxel is classified as an irritant and has a 27-hour stability. The usual dose range is 135 mg/m² to 250 mg/m² for patients with solid tumors and 200 mg/m² to 390 mg/m² for patients with leukemia.⁴

POTENTIAL TOXICITY OF PACLITAXEL

During Infusion

The primary toxicities that may occur during infusion of paclitaxel are hypersensitivity reactions (HSRs) and cardiotoxicity (Table 1). Diarrhea is a minor reaction that also can occur during paclitaxel infusions. Local reactions after paclitaxel infiltration include swelling, erythema, and mild pain.

Hypersensitivity reactions. It is unclear whether HSRs are related to paclitaxel or Cremophor. The mechanism for these reactions does not seem to be immunologically based but appears mediated by histamine release. Skin tests with paclitaxel have not been useful in predicting HSRs.⁵ The mechanisms of HSRs associated with paclitaxel seem analogous to those associated with iodinated radiocontrast dye. Therefore, the pre-

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0749-2081/93/0904-2004\$5.00/0*

Table 1. Nursing Care of the Patient Receiving Paclitaxel

High risk for injury (HSR)	1. Administer premedications as prescribed. 2. Assess vital signs as follows: 3-hour infusion—every 15 minutes for first hour, then hourly; 24-hour infusion—every 15 minutes for first hour, then every 4 hours. 3. Educate patient about potential HSRs and about starting premedications at home. Inform patient about signs and symptoms that should be reported to the nurse/physician.
High risk for altered cardiopulmonary tissue perfusion	1. Assess vital signs throughout infusion, including apical pulse. 2. Have pretreatment ECG; reassess if irregular heart rate noted.
Altered protection	1. Monitor pretreatment and follow-up CBCs for use of G-CSF in future cycles of paclitaxel. 2. Teach patient regarding neutropenic precautions. 3. Educate patient regarding signs of infection for which the physician should be notified. 4. Teach patient how to self-administer G-CSF, if indicated.
Pain/high risk for peripheral sensory/perceptual alteration	1. Educate patient regarding potential for discomfort related to arthralgias/myalgias and neurotoxicity. 2. Monitor level of symptoms and effectiveness of treatment for such.
Anxiety related to treatment/disease status	1. Allow patient to vent feelings regarding disease status and treatment. 2. Offer support to patient regarding investigational nature of treatment and disease status.

Abbreviations: CBC, complete blood count; ECG, electrocardiogram; G-CSF, granulocyte colony-stimulating factor. Adapted with permission.⁹

medications used (H_2 receptor antagonist, antihistamines, and corticosteroids) are consistent with those given prophylactically to reduce HSRs to radiographic contrast dyes.⁵ These premedications do not fully protect patients from HSRs but seem effective in reducing the incidence.

Early studies indicated that approximately 10% of patients receiving paclitaxel experience HSRs. However, only a few experience severe reactions.⁶ Approximately 50% of HSRs occur during the first 2 to 3 minutes of paclitaxel infusion, whereas 78% occur within 10 minutes of starting therapy.⁵ Most patients experience HSRs during the first two doses, whereas very few experience HSRs with later doses.⁵

Initially, HSRs appeared to be indirectly related to the duration of the paclitaxel infusion. HSRs were observed in 16% of patients given 3-hour infusions, 13% of patients given 6-hour infusions, and 7% of patients given 24-hour infusions.⁵ It is important to note that the majority of patients given 24-hour infusions had also received prophylactic medications. Based on this information, 24-hour infusions of paclitaxel were initially recom-

mended, along with the prophylactic premedications similar to those used for contrast dye reactions.⁵ Recently, shorter infusions (3- and 6-hour) have proven safe when administered in conjunction with the premedications.⁷ With the addition of the premedications, the rate of HSRs due to TAXOL have been less than 10%.⁶

Cardiotoxicity. Cardiac disturbances have been reported during paclitaxel infusions and usually occur later in the infusion.⁸ Asymptomatic bradycardia with rates of 30 to 50 beats per minute have been noted in approximately 30% of patients with no cardiac risk factors; this is not an indication for discontinuing paclitaxel.¹ However, a few patients have developed significant arrhythmias related to paclitaxel use, including atrioventricular block and third-degree block, and one patient was reported to have asystole for 7 seconds. Ventricular tachycardia also has been observed.¹ Several patients complained of atypical chest pains that may have been related to HSRs during paclitaxel infusions. A fatal myocardial infarction not preceded by arrhythmia occurred in one patient receiving paclitaxel.⁸ Although significant cardio-

toxicity is infrequent, close assessment of cardiac effects during the paclitaxel infusion is warranted (Table 1).

Diarrhea. Diarrhea has been reported as a potential side effect of paclitaxel. Diarrhea usually occurs during the second 12 hours of a 24-hour infusion, is generally mild, and rarely requires treatment.

Infiltration. Infiltrations of paclitaxel are managed with the application of heat as used in the management of vinca alkaloid extravasations. Alterations at the site of administration include erythema, tenderness and discomfort, and cellulitis in areas of infiltration. Although site reactions have been reported, peripheral administration of paclitaxel is rarely a problem. Therefore, central venous access is not required for administration.¹

After Infusion. After paclitaxel administration, patients may experience multiple side effects, including myelosuppression, nausea, mucositis, neurological toxic effects, myalgias/arthralgias, fatigue, headache, taste alterations, minor alterations in liver and renal function tests, and alopecia. Because these side effects occur primarily after discharge, patients must receive education about them and instructions to call if signs of infection occur or if myalgias/arthralgias are severe enough to require medical management.

PRETREATMENT PATIENT CARE

Before initiating the patient's paclitaxel infusion, the nurse must complete a head-to-toe assessment of the patient, with specific concentration on the patient's gastrointestinal functioning (ie, pre-existing nausea, vomiting, diarrhea, mucositis), signs and symptoms of infection, and preexisting neurotoxicity. Because many patients receiving paclitaxel have had previous cisplatin treatment, the level and character of preexisting neurotoxicity needs to be determined so that the effect of paclitaxel on the patient's neurological system will be known. A full set of vital signs are taken, as well as height and weight to verify body surface area and drug-dose calculation. In addition, the patient's venous status needs to be assessed for ability to withstand the infusion(s).

Because many of the HSRs to contrast dye and to paclitaxel are believed to have similar mechanisms of action, the patient's history of reactions to

other agents such as contrast dye must be determined.⁶ In addition, the patient's cardiac history should be ascertained.

Laboratory data also are important and should consist of a complete blood count and a full chemistry panel including liver function and renal function studies. A baseline electrocardiogram is necessary to fully assess any rhythm changes that may occur during paclitaxel infusion.

The patient's response to previous treatments with paclitaxel also must be assessed, including the level of myelosuppression, neurotoxicity, and myalgias/arthralgias, as well as any other symptoms the patient may have exhibited. In patients who have had significant myelosuppression, granulocyte colony-stimulating factor (G-CSF) is added to the treatment regimen.

During and after this assessment, patient education regarding treatment is initiated. Before starting the infusion, the nurse should discuss the method of drug administration, duration of the infusion, and when the patient will be able to leave the clinic/hospital. It is also important to convey the need for the patient to notify the nurse of any signs or symptoms of HSRs or cardiac arrhythmias.

Premedications

To prevent HSRs in patients receiving paclitaxel, the following premedications are recommended: (1) dexamethasone 20 mg orally or IV 12 and 6 hours before paclitaxel infusion; (2) diphenhydramine 50 mg IV 30 minutes before paclitaxel infusion; and (3) cimetidine 300 mg IV or ranitidine 50 mg IV 30 minutes before paclitaxel infusion.

Because dexamethasone is recommended 12 and 6 hours before treatment, patients are required to take this medication at home before paclitaxel infusion. In our clinical experience at Fox Chase Cancer Center, compliance in taking the dexamethasone before admission has been questionable. In addition, initiation of the paclitaxel infusion may be delayed, requiring retreatment with IV dexamethasone before initiating paclitaxel. Therefore, at Fox Chase Cancer Center, oral dosing with dexamethasone has been discontinued, and all patients now receive IV dexamethasone approximately 1 hour before beginning paclitaxel. This

change has caused no apparent increase in the incidence of HSRs in patients receiving paclitaxel.

ADMINISTRATION

Paclitaxel can be admixed into glass bottles or polyolefin or polypropylene bags.³ Glass bottles are less expensive than either of the bags. However, in clinical experience, the glass bottles periodically leak around the rubber sheath that covers the tubing insertion site approximately 2 to 3 hours after beginning the paclitaxel infusion. Bags avoid this problem and are less cumbersome than bottles.

Paclitaxel can be admixed in either 5% dextrose, 0.9% sodium chloride, 5% dextrose, and .9% sodium chloride, or 5% dextrose in Ringer's solution. Because the stability of paclitaxel is now determined to be 27 hours, one 24-hour bag or bottle of paclitaxel can be admixed. However, because there may be delays in initiating the infusion and because patients may have an adverse reaction requiring a delay in reinitiating or discontinuing the infusion, two 12-hour infusions offer the advantage of increased flexibility in the initiation of the infusion, both for the pharmacy and nursing staff workload and for minimized drug loss if the infusion is discontinued.

Polypropylene-lined administration sets (nitroglycerin tubing) are recommended for paclitaxel administration. Non-DEHP-containing Huber needles are available for use with central venous access ports; however, Huber needles containing small lengths of DEHP-containing tubing have been used successfully in the administration of paclitaxel. In addition, 0.22 μ m in-line filters are necessary to filter the minute particles found in the dilution of paclitaxel.³ The filters may require changing more frequently than every 24 hours because of "clogging" during the paclitaxel infusion.

PATIENT MONITORING

In the initial paclitaxel trials, patients underwent constant cardiac monitoring. Because the primary arrhythmia associated with paclitaxel is asymptomatic bradycardia, constant cardiac monitoring is no longer necessary. During the first 15 minutes of the infusion, the patient should be observed for possible HSRs. Because there is increased risk for HSRs early during the paclitaxel infusions, frequent assessment of pulse (apical), respirations,

and blood pressure every 15 minutes for the first hour of each infusion is necessary. Thereafter, apical pulse, respirations, and blood pressure are assessed every 4 hours throughout the infusion. Any irregularity in heart rate warrants a repeat ECG to assess possible arrhythmias.

MANAGEMENT OF TOXICITIES

If a severe HSR occurs, the paclitaxel infusion is stopped and the HSR is treated. Once a patient fully recovers from the reaction, a retreat of paclitaxel infusion may be given after readministration of the premedication regimen and reinitiation of paclitaxel at the previous or a reduced rate. No minimum time between trials of paclitaxel has been established. Because these reactions are primarily nonimmunologically based (ie, nonallergic), many patients who have them are subsequently able to receive paclitaxel without further reactions.

Cardiac arrhythmias can usually be detected through auscultation of an apical heart rate. Bradycardia (30 to 50 beats per minute) most commonly occurs while patients receiving paclitaxel are sleeping. Severe bradycardia (<50 beats per minute) can usually be reversed by awakening patients and having them walk a short distance. Other cardiac arrhythmias may necessitate discontinuation of paclitaxel.⁸

Gastrointestinal toxicities (ie, diarrhea and nausea) rarely occur while patients are receiving paclitaxel. Prochlorperazine may be given as needed to relieve nausea, but antiemetics are rarely needed on a scheduled basis with paclitaxel.

PSYCHOSOCIAL SUPPORT

Most patients receiving paclitaxel have tumors that have failed to respond to other chemotherapy agents. They are anxious about the potential effectiveness of this new treatment. Paclitaxel has received wide media coverage, which has also caused patients to seek it out—even for the management of disease sites for which paclitaxel may not yet be indicated. Patients believe that this new drug is the answer to their hopes for a cure. When patients fail to respond to paclitaxel and the drug is discontinued, they may become very anxious, and lack of effective treatment becomes an issue.

Health care professionals can assist patients in coping with their disease and paclitaxel treatment

by providing patient education and by helping them be realistic in their expectations (Table 1). Patients should be informed from the outset of the measures that will be used to assess response and to assess if and when therapy will be discontinued. Frequently, patients require time to absorb this information and to prepare for the possibility of treatment failure. Specific information regarding timing of the discontinuation of treatment is helpful in assisting the patient to cope with this event.

FOLLOW-UP/PATIENT EDUCATION

Patient education should focus on potential side effects of paclitaxel and should also include the hospital/clinic routine related to paclitaxel infusion and patient monitoring. Specific instructions should be provided regarding symptoms that need to be reported to the physician. In addition, side effects that might delay the next treatment, primarily myelosuppression, should be discussed. The patient should also be informed of the required follow-up assessment(s).

Patients who receive paclitaxel at doses above 190 mg/m² or in combination with other myelosuppressive drugs, or those who have had signifi-

cant myelosuppression after a previous paclitaxel treatment, also will receive G-CSF. These patients need instruction in how to inject this agent. Because a return-demonstration is usually not possible while the patient is an inpatient, home care follow-up to assess the patient's injection technique and to provide further education is often necessary.

CONCLUSION

Paclitaxel is a new antimicrotubule agent that has recently been approved by the FDA. Paclitaxel can be safely administered in the inpatient or outpatient setting. Nursing care is an important component in the safe administration of paclitaxel. Nurses provide direct patient care, which is important in the prevention and management of side effects, as well as patient education. Nursing care also includes psychosocial support not only for patients receiving paclitaxel but also for those requesting paclitaxel therapy and those whose tumors are no longer responding. As paclitaxel use is expanded, more information will be generated about the drug itself and the management of related side effects.

REFERENCES

- McGuire WP, Rowinsky EK, Rosenshein NB, et al: Taxol: A unique antineoplastic agent with significant activity in advanced ovarian epithelial neoplasms. *Ann Intern Med* 111: 273-279, 1989
- Lorenz W, Reimann H, Schmal A, et al: Histamine release in dogs by Cremophor EL and its derivatives: Oxyethylated oleic acid is the most effective constituent. *Agents Actions* 7:63-67, 1977
- Waugh W, Trissel L, Stella V: Stability, compatibility and plasticizer extraction of Taxol (NSC-125973) injection diluted in infusion solutions and stored in various containers. *Am J Hosp Pharm* 48:1520-1529, 1991
- Rowinsky EK, Burke P, Karp J, et al: Phase I and pharmacodynamic study of Taxol in refractory acute leukemias. *Cancer Res* 49:4640-4647, 1990
- Weiss R, Donehower R, Wiernik P, et al: Hypersensitivity reactions from Taxol. *J Clin Oncol* 8:1263-1268, 1990
- Weiss R: Hypersensitivity reactions. *Semin Oncol* 19(5): 459-477, 1992
- Swenerton K, Eisenhauer E, ten Bokkel Huinink W, et al: Taxol in relapsed ovarian cancer: High vs low dose and short vs long infusion. A European-Canadian study coordinated by the NCI Canada Clinical Trials Group. *Proc Am Soc Clin Oncol* 12:256, 1993 (abstr 810)
- Rowinsky EK, McGuire WP, Guarnieri T, et al: Cardiac disturbances during the administration of Taxol. *J Clin Oncol* 9:1704-1712, 1991
- Rogers B: Taxol: A promising new drug of the 90s. *Oncology Nursing Forum* 20:1483-1489, 1993